

An effective Gordan–Noether theorem for quaternary quintics

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Abstract

Let k have characteristic zero and let $f \in k[x_0, x_1, x_2, x_3]$ be homogeneous of degree five with $\det \text{Hess}(f) = 0$ and generic (function-field) Hessian rank three. We prove that the Hessian kernel contains a nonzero *constant* vector $e \in k^4$ – equivalently, f is a cone: after a linear change of coordinates it depends on three variables. This is the quintic, corank-one case of the classical Gordan–Noether theorem that quaternary forms with vanishing Hessian are cones; the proof given here is new, elementary, and effective. It runs entirely inside the polynomial ring: the unique polar relation of the gradient produces a primitive self-vanishing kernel field by a locally-nilpotent-derivation gcd descent; a degeneration-of-trajectories argument forces the projective image of that field to be a single line; and a first-order Wronskian elimination then forces the field to be constant or the rank to drop. Every algebraic identity in the proof is certified by an exact frozen checker.

The homogeneity hypothesis is not removable. We exhibit the explicit inhomogeneous potential

$$f = x(z^2 + z^4) + y(z^3 + z^5) + w^3$$

with layers of degrees 3, 4, 5, 6, whose Hessian has generic rank three and whose one-dimensional kernel field is *moving*: no nonzero constant kernel vector exists. The example is sharp in a precise sense: the polar relation, the primitive descent, and the self-vanishing identity all survive verbatim, and exactly the two consequences of Euler’s identity fail. A short independent argument – found by an adversarial audit inside the source campaign – isolates the same mechanism on the subclass of quintics linear in two variables, where Euler’s relation upgrades proportional gradients to a constant ratio in one line.

The theorem removes the last literature dependency from the homogeneous 3+4+5 stratification of an ongoing program on the four-variable Hessian conjecture HC(4), whose resolution would decide the planar Jacobian conjecture; that program remains open and nothing beyond the stated theorems is claimed here.

1 Introduction

Hesse claimed [5] that a projective hypersurface whose Hessian determinant vanishes identically is a cone. Gordan and Noether [4] proved this for forms in $n \leq 4$ variables and

*This manuscript was researched and written entirely by artificial-intelligence systems. The mathematics was produced during an autonomous multi-agent research campaign on the Jacobian-conjecture cascade (fleet repository 01-jc-cascade, 2026-08-04 through 2026-08-18); the two main results carry ledger identifiers H248 and H249, six recorded cross-window audit verdicts (five distinct auditors, all CONFIRMED), and two hostile referee rounds each. The present paper was organized and written from those records by Claude (Anthropic) on 2026-09-04. No human being wrote or verified the mathematical content. Section 8 states the verification model, every load-bearing citation, and an explicit unresolved priority-check caveat.

refuted it for every $n \geq 5$; the quaternary case has remained a small classic, with several modern treatments [3, 1]. This paper gives a new, self-contained, machine-verified proof of its quintic, corank-one case, together with a sharp inhomogeneous counterexample showing exactly which two consequences of homogeneity carry the theorem.

Theorem 1.1 (main theorem; ledger H248). *Let k be a field of characteristic zero and let $f \in k[x_0, x_1, x_2, x_3]$ be homogeneous of degree five with $\det \text{Hess}(f) = 0$ and $\text{rank}_{k(x)} \text{Hess}(f) = 3$. Then there is a nonzero constant vector $e \in k^4$ with*

$$\text{Hess}(f)e = 0,$$

equivalently $e \cdot \nabla f = 0$: after a linear change of coordinates f depends on three variables.

Theorem 1.2 (sharpness; ledger H249). *Over every characteristic-zero field, the polynomial*

$$f = x(z^2 + z^4) + y(z^3 + z^5) + w^3 \in k[x, y, z, w],$$

with nonzero homogeneous layers exactly in degrees 3, 4, 5, 6, satisfies $\det \text{Hess } f = 0$ and $\text{rank}_{k(x,y,z,w)} \text{Hess } f = 3$, yet its Hessian kernel contains no nonzero constant vector. Moreover the failure is located exactly: the kernel field produced by the polar-relation construction of Theorem 1.1 is still primitive and self-vanishing, and precisely the two Euler identities $v(v(X)) = 0$ and $v \cdot \nabla f = 0$ become false.

The pair is a closed answer to a well-posed question: *does the constant-kernel (cone) conclusion for degenerate quaternary Hessians persist with the hypotheses actually used?* Yes for one homogeneous quintic layer, by a proof that never leaves the polynomial ring; no for inhomogeneous multi-layer potentials, by an explicit certificate; and the boundary between the two is Euler’s identity and nothing else. A second, independent proof of the linear-in-two-variables subcase (Proposition 5.1) displays that boundary in one line.

Motivation

The question is not antiquarian. Writing HC(4) for the four-variable Hessian (Legendre-transform) conjecture, one has the implication $\text{HC}(4) \Rightarrow \text{JC}(2)$ – the planar Jacobian conjecture – via the symmetric reduction of de Bondt and van den Essen [2]; following the 2026 collapse of the higher-dimensional Jacobian landscape (an explicit non-injective Keller map on $\mathbb{C}^3$, and Meng–Yang’s five-variable refutation of the Hessian conjecture, leaving exactly $n = 4$ open [6]), the four-variable case became the live gateway. The source campaign stratifies Keller potentials with layers 3+4+5; every stratum theorem needed one upstream premise, namely a *constant* missing direction for the quintic layer. Under the campaign’s standing rule that no literature theorem may sit inside a proof chain, Theorem 1.1 supplies that premise in-house – it was, at campaign close, the most heavily audited single claim in the fleet ledger – and Theorem 1.2 shows the premise is genuinely about the homogeneous layer: it fails for the inhomogeneous total potential. HC(4) itself, and hence JC(2), remain open; nothing in this paper claims otherwise.

Novelty status, stated plainly

The *statement* of Theorem 1.1 is contained in the classical Gordan–Noether theorem and is not claimed as new. The contributions are the proof – elementary, effective, uniform over all characteristic-zero fields, with every identity exactly certified – and the sharp counterexample of Theorem 1.2, which we have not found in the literature. One priority check is outstanding: a 2014 article (Jin–Guo) on quaternary forms with vanishing Hessian could

not be obtained in full text during the campaign's audits, and the campaign's own rule – adopted here – is that novelty of the proof is not asserted until that article is read. Section 8 records this.

2 Conventions and the cone equivalence

Convention 2.1. k is a field of characteristic zero, $x = (x_0, x_1, x_2, x_3)$, $g = \nabla f$ is the gradient column, $C = \text{Hess}(f) = (\partial^2 f / \partial x_i \partial x_j)_{ij}$, and generic rank means rank over the rational function field $k(x)$. For a polynomial vector field $u = (u_0, \dots, u_3)$, $D_u = \sum_i u_i \partial_i$ denotes the associated derivation and $J(u)$ its Jacobian matrix, so that $D_u u = J(u)u$. A vector field is *primitive* if the gcd of its entries is 1.

Lemma 2.2 (cone equivalence). *Let f be homogeneous of degree $d \geq 2$ and $e \in k^4$ a nonzero constant vector. The following are equivalent: (i) $\text{Hess}(f)e = 0$; (ii) $e \cdot \nabla f = 0$; (iii) after a linear change of coordinates taking e to a coordinate direction, f does not involve that variable.*

Proof. $\text{Hess}(f)e = \nabla(e \cdot \nabla f)$, so (i) says $e \cdot \nabla f$ is a constant; being homogeneous of degree $d - 1 \geq 1$, it is then zero, giving (ii). Given (ii), extend e to a basis; in the new coordinates $\partial f / \partial x'_0 = e \cdot \nabla f = 0$, giving (iii). (iii) $\Rightarrow$ (i) is immediate. $\square$

3 Proof of the main theorem

Throughout this section f is as in Theorem 1.1. We may extend scalars to an algebraic closure $\bar{k}$ while proving the claim: generic rank is unchanged, and at the end the equation $\text{Hess}(f)e = 0$ for a constant e is a finite homogeneous *linear* system in the entries of e with coefficients in k ; a nonzero solution over $\bar{k}$ forces a nonzero kernel over k . So assume k algebraically closed.

3.1 A primitive self-vanishing kernel field

If the four entries of g are linearly dependent over k , a constant relation already gives the theorem via Lemma 2.2. Assume they are independent.

Differential rank equals transcendence degree. For elements $h_1, \dots, h_N$ of a rational function field over k , the rank of $dh_1, \dots, dh_N$ equals $\text{trdeg}_k k(h_1, \dots, h_N)$: choose a transcendence basis among the h_i ; its differentials are independent (extend to a transcendence basis of the ambient field, over which the residual algebraic extension is separable in characteristic zero, and differentiate); every remaining h_j is separably algebraic over the basis, and differentiating its minimal polynomial expresses dh_j in the span. This proves both inequalities with no geometric input.

Applying this to the entries of g , whose Jacobian is C : a nonzero 3×3 minor of C gives rank at least three, and $\det C = 0$ gives at most three, so $\text{trdeg}_k k[g_0, \dots, g_3] = 3$. Hence the kernel of

$$k[Y_0, Y_1, Y_2, Y_3] \longrightarrow k[g_0, g_1, g_2, g_3]$$

is a height-one homogeneous prime; the polynomial ring being a UFD, it is generated by one irreducible homogeneous G (an irreducible factor of any nonzero element generates a height-one prime inside it, and strict containment of height-one primes is impossible). Put

$$w = \nabla G(g).$$

Then $w \neq 0$: otherwise G would divide each of its own first partials, which have smaller degree, forcing them all to vanish – impossible for a nonconstant polynomial in characteristic zero. Differentiating $G(g) = 0$ by the chain rule,

$$Cw = 0, \quad J(w) = \text{Hess}(G)(g) C, \quad \text{hence} \quad D_w w = J(w) w = 0. \quad (2)$$

Primitive descent. Write $w = \rho v$ with $\rho = \gcd$ of the entries and v primitive. We claim

$$D_v v = 0. \quad (3)$$

By (2) the derivation D_w kills each of its own coefficients, hence is locally nilpotent (its square vanishes on the generators, and degree bounds the iterate needed on any polynomial). Let π be an irreducible factor of ρ and pick an entry v_i not divisible by π ; the π -adic leading term of $D_w(\rho v_i) = 0$ gives $\pi \mid D_w \pi$. But a locally nilpotent derivation admits no nonzero Darboux cofactor: if $D_w \pi = A\pi$ then $\exp(tD_w)(\pi) = U(t)\pi$ with U invertible in $k[t, x]$ and $U(0) = 1$, so $U = 1$ and $D_w \pi = 0$. Hence $D_w \rho = 0$, and $D_w = \rho D_v$ turns (2) into $\rho^2 D_v v = 0$, proving (3).

Self-vanishing. All entries of v are homogeneous of one degree m ; if $m = 0$ we are done, so assume $m > 0$. Iterating (3), the derivation D_v kills v , whence the exact transport identity

$$v(x + t v(x)) = v(x) \quad (4)$$

in $k[t, x]$. Comparing coefficients of t^m gives

$$v(v(x)) = 0. \quad (5)$$

Thus the projective image Y of v lies in the base locus $B = V(v_0, \dots, v_3)$; primitivity excludes a common hypersurface component, so $\text{codim } B \geq 2$ and $\dim Y \leq 1$. Finally, along a transported kernel line, $C(x + tv)v = 0$; the matrix C is cubic, its t^3 coefficient is $C(v)v$, and quintic Euler gives

$$C(v)v = 4g(v) = 0, \quad (6)$$

the transported cubic coefficient used below (this, and every displayed identity of this section, is verified exactly by the frozen checker; see Section 6).

3.2 The degenerate trajectories force a line image

If $\dim Y = 0$ then v has constant projective direction and the theorem follows from $Cv = 0$. Suppose $\dim Y = 1$; we show Y is a projective line.

Let $U = \mathbf{P}^3 \setminus B$ and $\psi : U \rightarrow Y$ the map defined by v ; a general fibre has dimension two. Fix an irreducible surface component of a fibre closure, F_q ; every surface in $\mathbf{P}^3$ is a hypersurface, and the transport identity (4) says that for every defined $x \in F_q$ the line $\langle x, q \rangle$ lies in $\overline{F_q}$.

Choose two distinct general points $q_1, q_2 \in Y$. The hypersurfaces F_{q_1}, F_{q_2} meet in dimension at least one (two homogeneous equations have height at most two, the irrelevant ideal height four), and their intersection lies in B , since a point of U cannot take both values. Choose a point $t \neq q_1, q_2$ on a curve of that intersection. For each i , F_{q_i} has local dimension two at t while $\dim B \leq 1$, so some curve through t in F_{q_i} is not contained in B ; normalize it at a point over t and lift to a formal arc of affine-cone representatives $x_i(\epsilon) \in \bar{k}[[\epsilon]]^4$ with $x_i(0) = T \neq 0$. On the punctured generic point the projective value is q_i , so for a fixed representative Q_i ,

$$v(x_i(\epsilon)) = c_i(\epsilon) Q_i, \quad c_i(\epsilon) \neq 0, \quad c_i(0) = 0.$$

Apply (4) over $\bar{k}(\epsilon)$ with transport parameter $\lambda/c_i(\epsilon)$ and specialize at $\epsilon = 0$:

$$v(T + \lambda Q_i) = 0 \quad \text{for every } \lambda.$$

Thus each line $\langle t, q_i \rangle$ lies in B . A general point of the irreducible curve Y lies on no other irreducible curve component of B , so $\langle t, q_i \rangle = Y$: Y is a projective line. (This is the step that absorbs the trajectories on which the rational kernel chart degenerates: they cannot support a separate moving-kernel component.)

After a constant linear change of source coordinates (r, p, s, z) placing this line as the (s, z) -plane direction,

$$v = (0, 0, a, b), \quad a, b \in k[r, p, s, z]. \quad (7)$$

We remove the fibre variables. Work over $K = k(r, p)$; primitivity gives $\gcd(a, b) = 1$ in $K[s, z]$, and (3) reads

$$a a_s + b a_z = 0, \quad a b_s + b b_z = 0. \quad (8)$$

If $b = 0$ then $a \in K^\times$. Otherwise the second equation with coprimality gives $b \mid b_s$, so $b_s = 0$ by degree, then $b_z = 0$, so $b \in K^\times$. In the first equation let $m = \deg_s a$: for $m \geq 2$ the term $a a_s$ has s -degree $2m - 1$ against at most m for $b a_z$; for $m = 1$, writing $a = A(z)s + B(z)$, the s -coefficient reads $A^2 + b A' = 0$, and z -degree forces $A = 0$. Hence $m = 0$ and then $a_z = 0$ by (8). Clearing a common denominator,

$$v = (0, 0, a(r, p), b(r, p)), \quad \gcd(a, b) = 1. \quad (9)$$

3.3 The first-order group forces constancy

Quintic Euler and $Cv = 0$ give $v \cdot g = 0$. Differentiating in r and p , symmetry of C removes the second-order terms:

$$v_r \cdot g = 0, \quad v_p \cdot g = 0.$$

With (9) the complete group is

$$a f_s + b f_z = 0, \quad a_r f_s + b_r f_z = 0, \quad a_p f_s + b_p f_z = 0. \quad (10)$$

(These are exactly the connection terms retained because the kernel coordinate itself moves: on the chart $b = 1$, the change $X(r, p, s, z) = (r, p, s + a(r, p)z, z)$ has $X_z = v$ and $0 = (f \circ X)_{rz} = X_r^\top C X_z + g \cdot v_r = g \cdot v_r$, and likewise in p . This is the group a silent constancy assumption would lose.) Setting

$$W_r = a b_r - b a_r, \quad W_p = a b_p - b a_p,$$

elimination from (10) gives

$$W_r f_s = W_r f_z = W_p f_s = W_p f_z = 0. \quad (11)$$

If some Wronskian is nonzero then $f_s = f_z = 0$ identically, and $\text{Hess}(f)$ has rank at most two – against the rank-three hypothesis. So $W_r = W_p = 0$. If $a \neq 0$, then b/a has both partial derivatives zero in $k(r, p)$: for the reduced fraction A/B the r -equation makes $A \mid A_r$ and $B \mid B_r$, so both vanish by degree, and likewise in p ; hence $b/a \in k$. The coprime pair (a, b) is therefore constant up to a unit (the cases $a = 0$ or $b = 0$ being immediate), so v has a nonzero constant projective direction e , and $Cv = 0$ gives $Ce = 0$ identically. $\square$

By Lemma 2.2, f is a cone. This completes Theorem 1.1.

4 Sharpness: the moving-kernel counterexample

Proof of Theorem 1.2. Put $A = z^2 + z^4$, $B = z^3 + z^5 = zA$, and $R = xA'' + yB''$. Then

$$C = \text{Hess } f = \begin{pmatrix} 0 & 0 & A' & 0 \\ 0 & 0 & B' & 0 \\ A' & B' & R & 0 \\ 0 & 0 & 0 & 6w \end{pmatrix}.$$

The first two rows are proportional over $k(z)$, so $\det C = 0$; the minor on rows and columns (x, z, w) equals $-6w(A')^2 = -24wz^2(1 + 2z^2)^2 \neq 0$, so the generic rank is three. The kernel is generated by $\kappa = (B', -A', 0, 0)$, with primitive form (after removing the gcd z)

$$v = (z(3 + 5z^2), -2(1 + 2z^2), 0, 0).$$

No nonzero constant e satisfies $Ce = 0$: the x -row gives $A'e_z = 0$ so $e_z = 0$; the w -row gives $e_w = 0$; the z -row then reads $(2z + 4z^3)e_x + (3z^2 + 5z^4)e_y = 0$, and the z, z^2 coefficients force $e_x = e_y = 0$.

The construction of Section 3.1 survives verbatim. The gradient is $g = (A, B, xA' + yB', 3w^2)$, of transcendence degree three, and the unique height-one polar relation is generated by the irreducible

$$G(Y_0, Y_1) = Y_0^5 - Y_0^2 Y_1^2 - Y_1^4$$

(localizing at Y_0 and setting $t = Y_1/Y_0$ gives $G = Y_0^4(Y_0 - t^2 - t^4)$, whose localized quotient $k[t, (t^2 + t^4)^{-1}]$ is a domain, and $Y_0 \nmid G$). Pulling back, $\nabla G(g) = A^3 z v$: the polar construction produces exactly the moving field v , and since v has only x, y components with z -only coefficients,

$$D_v v = 0, \quad v(X + \tau v(X)) = v(X)$$

– primitive gcd descent and self-vanishing both hold. What fails is homogeneity, in exactly two places. First, the top- τ extraction (5): here

$$v(v(X)) = (0, -2, 0, 0) \neq 0,$$

and indeed the base locus is *empty*, by the Bézout identity $-4z v_1 + (-5z^2 - \frac{1}{2}) v_2 = 1$ – so the projective line-image argument of Section 3.2 has no source to run on. Second, the Euler consequence $v \cdot \nabla f = 0$ used to open Section 3.3: here

$$v \cdot \nabla f = z^3(1 + z^2)^2 \neq 0,$$

while the corresponding z -Wronskian $v_1 \partial_z v_2 - v_2 \partial_z v_1 = 2(3 + 9z^2 + 10z^4)$ is nonzero with no contradiction, because the first equation of the system (10) is precisely the false identity above. Thus the extension of Theorem 1.1 from one homogeneous quintic layer to the inhomogeneous $3+4+5+6$ sum is false, and it breaks at Euler's identity and nowhere else. $\square$

5 The Euler mechanism in one line: an independent subcase proof

The following proposition and proof arose as an adversarial *audit* of Theorem 1.1 inside the source campaign: the auditor, working without access to the authoring repository, tried to break the theorem by homogenizing Theorem 1.2's counterexample – and proved instead that this cannot work. It is included because it displays the load-bearing role of Euler's identity with maximal clarity.

Proposition 5.1. *Let $A, B \in k[z, w]$ be quartic forms, not both zero, and let $f = xA + yB$, a quaternary quintic linear in (x, y) . If $\text{rank}_{k(x)} \text{Hess } f = 3$ then $\text{Hess } f$ has a nonzero constant kernel vector; explicitly $B = \lambda A$ for a constant λ (or symmetrically), $f = (x + \lambda y)A(z, w)$ depends on three variables after a linear change, and the constant kernel direction is $(\lambda, -1, 0, 0)$.*

Proof. In coordinates (x, y, z, w) ,

$$\text{Hess } f = \begin{pmatrix} 0 & M \\ M^\top & N \end{pmatrix}, \quad M = \begin{pmatrix} A_z & A_w \\ B_z & B_w \end{pmatrix},$$

for some symmetric N , and Laplace expansion along the first two rows gives $\det \text{Hess } f = (\det M)^2 = J^2$ with $J = A_z B_w - A_w B_z$. Rank three forces $J \equiv 0$, i.e. $(B_z, B_w) = \lambda (A_z, A_w)$ for some rational λ . Euler’s identity for quartic forms gives

$$4B = zB_z + wB_w = \lambda(zA_z + wA_w) = 4\lambda A,$$

so $B = \lambda A$; differentiating, $B_z = \lambda_z A + \lambda A_z = \lambda A_z$ forces $\lambda_z A = 0$, and likewise $\lambda_w A = 0$, so λ is constant. Then $f = (x + \lambda y)A$, and $\text{Hess}(f)(\lambda, -1, 0, 0)^\top = \nabla((\lambda \partial_x - \partial_y)f) = \nabla(\lambda A - \lambda A) = 0$. $\square$

Remark 5.2. Theorem 1.2’s counterexample is exactly this shape with the quartic forms replaced by the *inhomogeneous* univariate polynomials $A = z^2 + z^4$, $B = zA$: there the ratio $B/A = z$ is a nonconstant function, and no Euler identity is available to force it constant. Homogeneity, through Euler’s relation, upgrades “gradients proportional” to “functions proportional by a constant”; that one upgrade is the entire distance between Theorems 1.1 and 1.2.

6 Effectivity and machine verification

The proof of Theorem 1.1 is effective: given f , the polar relation G is computable by elimination, the kernel field v by differentiation and gcd removal, and the final constant direction by linear algebra. Every displayed identity is certified by the frozen exact checker `check_hc4_three_layer_rank3_top_kernel_constancy.py`: the transported cubic coefficient (6), all four Wronskian eliminations (11), a rank-three constant-direction control, and a negative control (imposing the moving field $v = (0, 0, r, p)$ on a generic homogeneous quintic leaves exactly the six binary quintics in (r, p) , all of Hessian rank at most two). A separate exact search over coprime moving fields of degrees one to three returned zero non-binary survivors (`data/hc4_three_layer_rank3_moving_top_kernel_search.json`); this search is falsification evidence only – the theorem rests on Section 3. Theorem 1.2 is certified by `check_hc4_weighted_multilayer_rank3_moving_kernel.py`. In the source ledger the two results underwent two hostile referee rounds each and carry six recorded cross-window audit verdicts by five distinct auditors, all CONFIRMED; one auditor’s verdict is CONFIRMED-WITH-NAMED-SCOPE, its content being Proposition 5.1 together with the explicit statement that the general (nonlinear in (x, y)) case was confirmed only against attack, not re-derived – a scope the present paper’s Section 3 closes with the full argument. No step has been formalized in a proof assistant.

7 Application: the HC(4) stratification

In the source campaign’s stratification of four-variable Keller potentials $Q + H_3 + H_4 + H_5$, every branch theorem for the generic rank-three quintic layer (ledger H242–H246) required a

constant missing direction for that layer. Theorem 1.1 supplies the premise for *every* generic rank-three homogeneous quintic layer, removing the campaign’s last upstream literature dependency in that stratification, while Theorem 1.2 shows the premise cannot be obtained for the total inhomogeneous potential and must indeed be extracted layer-by-layer. The unrestricted four-variable Hessian conjecture, and with it the planar Jacobian conjecture, remain open; the shortest unresolved case in the campaign’s ledger is a two-layer 3+4 weighted determinant identity.

8 Provenance and disclosure

The mathematics was produced autonomously by an AI research fleet (repository 01-jc-cascade, 2026-08-04 through 2026-08-18) under a standing rule forbidding literature theorems inside proof chains; Theorem 1.1 exists precisely because the classical Gordan–Noether theorem could therefore not be cited. Classical and modern references on vanishing Hessians – Hesse [5], Gordan–Noether [4], Garbagnati–Repetto [3], Ciliberto–Russo–Simis [1] – are cited for context and priority, not used in any proof. The motivation chain (HC(4) $\Rightarrow$ JC(2), and the 2026 status of the cascade) cites [2, 6] and is likewise not load-bearing. **Priority caveat:** a 2014 article by Jin and Guo on quaternary forms with vanishing Hessian was inaccessible in full text during the campaign (repository record: CNKI retrieval failed); until it is read, novelty is claimed for nothing beyond what Section 6 documents – the specific proof, its effectivity and certification, and the counterexample – and even that subject to the check. All certificates and the audit trail are archived in the campaign repository.

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